# Sérgio Alexandre

**Mid-level Full-Stack Software Engineer · TypeScript, Next.js, NestJS, PostgreSQL, Applied AI**

Londrina, PR, Brazil · (43) 9 8873 2020 · sergioalexandre0716@gmail.com
linkedin.com/in/serg-alexandre · github.com/Serg-Ale · saerix.vercel.app/en/dev

## SUMMARY

Full-stack software engineer with 4+ years of experience, mid-level since Mar 2026. I build production web products end to end: Next.js and React front ends, REST APIs in NestJS and Node.js, PostgreSQL, Pix billing, OTP authentication with roles, analytics dashboards and AI agents with tool calling on WhatsApp. I care about quality (automated tests with over 80% coverage, CI/CD, pre-commit hooks), architecture and product decisions with the PO, and mentoring developers.

## EXPERIENCE

**Aeon Tech (formerly Union Audio)** · Software Engineer (Mid-level) — Mar 2026 – Present

- Proposed and led, with team and PO validation, **Corta-ai**, a SaaS for barbershops (Next.js, NestJS, PostgreSQL, n8n) now in pilot with two barbershops: schedule, services and cancellations through the web admin and WhatsApp.
- Implemented OTP login with roles per barbershop (multi-tenant, RBAC), daily Pix billing with Asaas webhooks and idempotent processing (trial, grace period and block) and analytics dashboards by plan.
- Designed the architecture: one NestJS API with two entry points (web admin and chat) behind a BFF, TypeORM migrations and e2e tests (Jest, Supertest) for business rules: double booking, multi-tenancy and cancellation deadlines.
- Built the MVP's WhatsApp booking agent: an n8n AI Agent (LangChain) with an LLM and **tool calling** (list services and barbers, check availability, create appointment) over the NestJS API; evolved the chat into fixed flows with human fallback, on a self-hosted Evolution API (cheaper than Meta's).
- Own product quality: unit and e2e tests (over 80% coverage), pre-commit hooks and CI/CD; take part in prioritization, QA and post-launch monitoring. Mentor an intern through a structured development plan and technical 1:1s.

*Junior Developer* — Jan 2024 – Mar 2026

- Delivered the product's front end (Next.js, React, TypeScript, Tailwind, GSAP) in its rebuild, in a team of 4–5: profile page and feedback modules with a SoundCloud player.
- Built the Aeon Audio landing page, with GSAP animation and kinetic text (aeonaudio.com.br).
- Architected the WhatsApp automation base (self-hosted Evolution API, n8n) with phone-number OTP login: sign-up and booking 24h from a phone, at a lower cost than Meta's official API.

**Kaizen** · Developer Partner — May 2026 – Present

- Equity partner at a smart-condominium startup, on **Atrium** (face access for guests), with a team across Madrid, Mexico City and Brazil.
- Own the flow from registering a person to the door opening, with temporary access and automatic events, in NestJS, Next.js and Prisma; integrate Hikvision and Control iD readers, plus the backoffice and WhatsApp flows.

**Tata Consultancy Services** · Analyst — Feb 2023 – Nov 2023

- Provided on-call support for the overnight critical systems of a large retail chain: maintained Java/Spring services (Maven/Gradle, JUnit, Mockito), monitored with Zabbix, with tickets in Jira Service Management, diagnosing and resolving incidents.
- Acted as the technical link between engineering and managers in high-severity incident war rooms.

*Intern* — Jun 2022 – Feb 2023

- First developer role: front end and back end (Java, Spring Boot, Node.js), front-end tests and Scrum.

## PROJECTS

- **job-agent** (personal, TypeScript, private code): an AI agent that discovers and scores job posts and fills applications with Playwright. The LLM (Claude, Anthropic API) only classifies fields and drafts answers, with **structured output** (JSON schema) checked against a closed list; it never sees identity data or controls the browser (least privilege), treats the posting text as untrusted input (**prompt injection** defense) and requires **human approval** before sending. Vitest, SQLite.
- **gym-stuff** (offline-first workout PWA, github.com/Serg-Ale/gym-stuff), **matrix-rain** (Python, public spec, tickets and TDD, github.com/Serg-Ale/matrix-rain) and **lightsync** (Python: a local service that syncs a smart bulb to the screen and audio; numpy, Textual, pytest).

## SKILLS

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| **Languages** | TypeScript, JavaScript, Python, SQL · **Familiar with:** FastAPI, Django |
| **Front end** | React, Next.js, Tailwind CSS, React Query, GSAP, HTML, CSS |
| **Back end** | Node.js, NestJS, REST APIs, system design, BFF, multi-tenancy, RBAC, webhooks, tRPC, Prisma, TypeORM, Java (Spring Boot) |
| **Data** | PostgreSQL, data modeling, migrations, Supabase |
| **Testing** | unit, integration and e2e testing, Jest, Supertest, Vitest, JUnit, Mockito, Playwright, TDD |
| **DevOps** | CI/CD, Docker, Git, Linux, Zabbix · **Practices:** Scrum, technical decisions, mentoring |
| **Applied AI** | LLMs (Anthropic API, Claude), tool calling, AI agents, structured output (JSON schema), prompt engineering, prompt injection defense, human-in-the-loop, n8n AI Agent (LangChain) |
| **AI-assisted dev** | Claude Code (skills, sub-agents, MCP), spec-driven development, TDD · **Automation:** n8n, WhatsApp / Evolution API |

## EDUCATION AND LANGUAGES

Systems Analysis and Development — UTFPR · Portuguese (native) · English (EF SET 72/100) — Jul 2022 – In progress